



COMSATS University Islamabad

Department of Computer Science

Course Information

Course Code: **CSC354**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Machine Learning**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

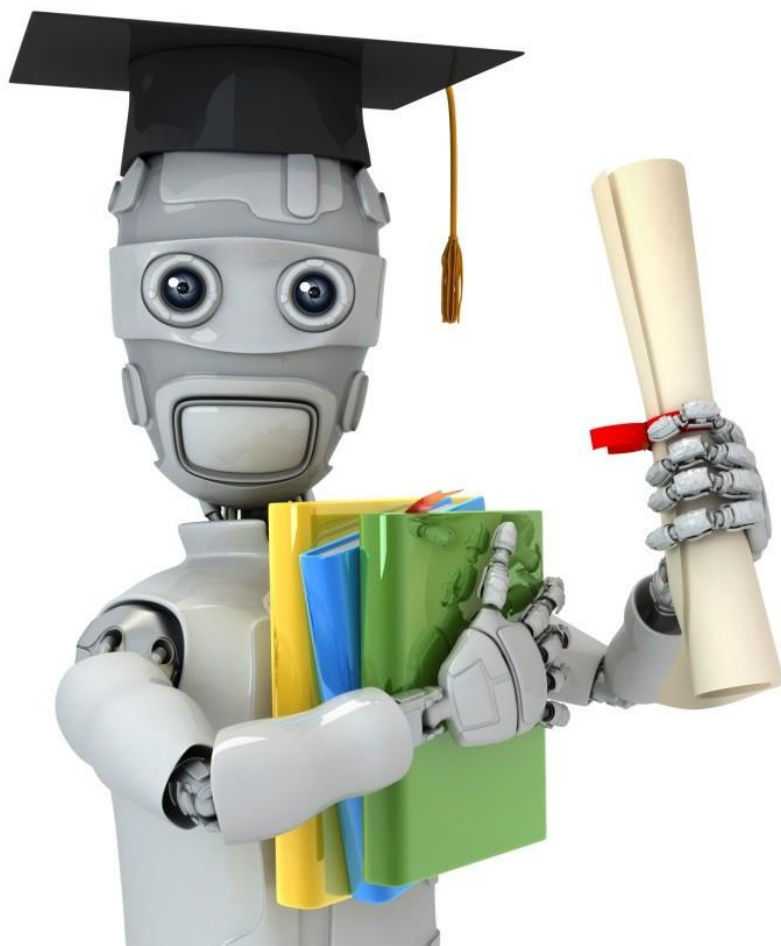
Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O'Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.



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- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on zeenatzulfiqar@cuiatd.edu.pk



Machine Learning

Introduction

Unsupervised
Learning Model
Self-Organizing
Maps (SOM)

Self-Organizing Maps (SOM)

- **Unsupervised Learning Algorithm**

SOM is an **unsupervised neural network** that learns patterns from unlabeled data without requiring target outputs

- **Developed by Teuvo Kohonen**

SOM, also known as the **Kohonen Map**,

- **Competitive Learning**

SOM uses **competitive learning**, where neurons compete to become the **Best Matching Unit (BMU)** for each input.

- **Dimensionality Reduction**

SOM transforms high-dimensional data into a **two-dimensional map**, making complex datasets easier to visualize.

Self-Organizing Maps (SOM)

- **Common Applications**

SOM is widely used for **clustering, customer segmentation, image processing, pattern recognition, anomaly detection, medical diagnosis, and data visualization.**

- **five keywords for SOM:**

- **Unsupervised Learning**

- **Competitive Learning**

- **Best Matching Unit (BMU)**

- **Dimensionality Reduction**

Self-Organizing Maps (SOM)

- Initialize the SOM with random weights.
- Present an input vector to the network.
- Calculate the Euclidean distance between the input and each neuron.
- Select the neuron with the smallest distance as the Best Matching Unit (BMU).
- Update the weights of the BMU and its neighboring neurons.
- Decrease the learning rate and neighborhood radius.
- Repeat the process until the network converges.
- Use the trained SOM to visualize and cluster similar data.

Example1

Question

A Self-Organizing Map (SOM) has **three neurons** with the following initial weight vectors:

Neuron	Weight Vector
N1	(0.3, 0.2)
N2	(0.6, 0.7)
N3	(0.8, 0.4)

Example1

- An input vector is:

$$X = (0.5, 0.6)$$

- The learning rate is:

$$\alpha = 0.4$$

Tasks:

- Calculate the Euclidean distance between the input vector and each neuron.
- Find the Best Matching Unit (BMU).
- Update the weight vector of the BMU.

Solution: Step 1: Write the Given Data

Input Vector:

$$X = (0.5, 0.6)$$

Learning Rate:

$$\alpha = 0.4$$

Initial Weights:

Neuron	Weight
N1	(0.3, 0.2)
N2	(0.6, 0.7)
N3	(0.8, 0.4)

Step 2: Calculate the Euclidean Distance

- The Euclidean distance formula is:

- $D = \sqrt{(x_1 - w_1)^2 + (x_2 - w_2)^2}$

- **Distance for N1**

- $D_1 = \sqrt{(0.5 - 0.3)^2 + (0.6 - 0.2)^2}$

- $= \sqrt{(0.2)^2 + (0.4)^2}$

- $= \sqrt{0.04 + 0.16}$

- $= \sqrt{0.20}$

- $D_1 = 0.447$

Step 2: Calculate the Euclidean Distance

- The Euclidean distance formula is:

- $D = \sqrt{(x_1 - w_1)^2 + (x_2 - w_2)^2}$

- **Distance for N2**

- $D_2 = \sqrt{(0.5 - 0.6)^2 + (0.6 - 0.7)^2}$

- $= \sqrt{(-0.1)^2 + (-0.1)^2}$

- $= \sqrt{0.01 + 0.01}$

- $= \sqrt{0.02}$

- $D_2 = 0.141$

Distance for N3

- $D_3 = \sqrt{(0.5 - 0.8)^2 + (0.6 - 0.4)^2}$
- $= \sqrt{(-0.3)^2 + (0.2)^2}$
- $= \sqrt{0.09 + 0.04}$
- $= \sqrt{0.13}$
- $D_3 = 0.361$

Step 3: Find the Best Matching Unit (BMU)

Compare the distances:
The smallest distance is **0.141**.
Therefore, N2 is the Best Matching Unit (BMU).

Neuron	Distance
N1	0.447
N2	0.141
N3	0.361

Step 4: Update the BMU Weight

The SOM weight update formula is:

- $W_{new} = W_{old} + \alpha(X - W_{old})$
- **Update the First Weight**
- Old weight = 0.6
- Input = 0.5

Learning rate = 0.4

- $W_1 = 0.6 + 0.4(0.5 - 0.6)$
- $= 0.6 + 0.4(-0.1)$
- $= 0.6 - 0.04$
- $= 0.56$

Update the Second Weight

- Old weight = 0.7
- Input = 0.6
- $W_2 = 0.7 + 0.4(0.6 - 0.7)$
- $= 0.7 + 0.4(-0.1)$
- $= 0.7 - 0.04$
- $= 0.66$

Step 5: Write the Updated Weight Vector

- the updated weight vector of the BMU (N2) is:
- $(0.56, 0.66)$

- Best Matching Unit (BMU): N2
- Updated Weight Vector of N2: (0.56, 0.66)

Neuron	Distance
N1	0.447
N2	0.141
N3	0.361

Format

- **Write the given input vector, weight vectors, and learning rate. Calculate the Euclidean distance for each neuron. Choose the neuron with the smallest distance as the Best Matching Unit (BMU).**
- **Update the BMU's weights using the formula:**
- $W_{new} = W_{old} + \alpha(X - W_{old})$
- **Write the final updated weight vector and identify the BMU.**

Self-Organizing Maps (SOM) Using a Real Dataset Example

Question

- A shopping mall wants to group customers with similar spending behavior using a **Self-Organizing Map (SOM)**.
- The dataset contains the following customer information:

Customer	Annual Income (×1000 \$)	Spending Score
C1	25	30
C2	40	45
C3	60	65
C4	80	85

Self-Organizing Maps (SOM) Using a Real Dataset Example

- Assume the SOM has **3 neurons** with the following initial weights:

Neuron	Weight Vector
N1	(20,25)
N2	(45,50)
N3	(75,80)

Take the first customer **C2 = (40,45)** as the input vector.

Learning rate:

$$\alpha = 0.5$$

Tasks

1. Calculate the Euclidean distance.
2. Find the Best Matching Unit (BMU).
3. Update the BMU weights.

Step 1: Draw the SOM

- Self-Organizing Map

- | | | | |
|---------|---------|---------|--|
| | | | |
| N1 | N2 | N3 | |
| (20,25) | (45,50) | (75,80) | |

- Input Customer

- $C2 = (40,45)$

Step 2: Calculate Euclidean Distance

- Formula

- $D = \sqrt{(x_1 - w_1)^2 + (x_2 - w_2)^2}$

- **Distance to N1**

- $= \sqrt{(40 - 20)^2 + (45 - 25)^2}$

- $= \sqrt{20^2 + 20^2}$

- $= \sqrt{400 + 400}$

- $= \sqrt{800}$

- $= 28.28$

- **Distance to N2**

- $= \sqrt{(40 - 45)^2 + (45 - 50)^2}$

- $= \sqrt{(-5)^2 + (-5)^2}$

- $= \sqrt{25 + 25}$

- $= \sqrt{50}$

- $= 7.07$

- **Distance to N3**

- $= \sqrt{(40 - 75)^2 + (45 - 80)^2}$

- $= \sqrt{35^2 + 35^2}$

- $= \sqrt{1225 + 1225}$

- $= \sqrt{2450}$

- $= 49.50$

Step 3: Compare Distances

Neuron	Distance
N1	28.28
N2	7.07
N3	49.50

- The **smallest distance is 7.07**, so

Step 4: Update the BMU Weight

- Formula
- $W_{new} = W_{old} + \alpha(X - W_{old})$
- Old Weight
- (45, 50)
- Input
- (40, 45)
- Learning Rate
- 0.5

- **First Weight**

- $45 + 0.5(40 - 45)$

- $= 45 - 2.5$

- $= 42.5$

- **Second Weight**

- $50 + 0.5(45 - 50)$

- $= 50 - 2.5$

- $= 47.5$

Step 5: Updated SOM

- Updated SOM

- +-----+=====+-----+
- | N1 || N2 ✓ || N3 |
- | (20,25) || (42.5,47.5) || (75,80) |
- +-----+=====+-----+

Neuron	Distance
N1	28.28
N2	7.07
N3	49.50

**Best Matching Unit (BMU): N2 Updated Weight of N2:
(42.5, 47.5)**